

Tanmoy Debnath

Backend Developer | Scalable Distributed Systems
Specializing in Microservices & Production Infrastructure
Techno College of Engineering Agartala

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SUMMARY

Backend Developer specializing in scalable distributed systems, RESTful API architecture, and high-performance microservices. Experienced in designing production-grade infrastructures using FastAPI, PostgreSQL, Redis, and event-driven architectures with Celery/RabbitMQ. Expert in system design, API security (JWT/RBAC), CI/CD automation, and cloud-native deployment.

EXPERIENCE

•Backend Developer & Automation Engineer

Sep 2025 – Present

Yupcha

Onsite

- Designed and deployed production-grade CI/CD pipelines using GitHub Actions and Docker, implementing blue-green deployment strategies and containerized microservices architecture to achieve zero-downtime releases.
- Designed and optimized RESTful APIs following OpenAPI standards using FastAPI, implementing PostgreSQL indexing, query optimization, caching strategies (Redis), and API rate limiting to handle 50k+ daily requests.
- Engineered high-availability backend systems with load balancing, health checks, and distributed service monitoring, ensuring 99.9% uptime and horizontal scalability under high traffic loads.
- Implemented JWT-based authentication and role-based access control (RBAC) to secure RESTful APIs and enforce fine-grained authorization policies across distributed services.
- Developed comprehensive unit and integration tests using PyTest to ensure backend reliability, improving code coverage by 30% and significantly reducing production defects.

PROJECTS

•Blockchain Certificate Verifier (Cyphire)

2026

Immutable platform for digitally-signed credentials anchored on the Polygon blockchain.

🐙 GitHub

- Developed backend services integrating FastAPI with Web3.py to execute secure blockchain transactions, implementing asynchronous processing, transaction validation, and fault-tolerant API design.
- Documented REST APIs using Swagger/OpenAPI specifications to standardize service contracts and improve developer onboarding; synchronized PostgreSQL state with blockchain consensus for data integrity.

•Link Ingestor API

2025

Distributed autonomous system for large-scale webpage data ingestion and async processing.

🐙 GitHub

- Architected event-driven microservices using Celery, Redis, and RabbitMQ to process 200+ concurrent tasks, implementing distributed task queues and horizontal worker scaling.
- Implemented retry mechanisms, exponential backoff strategies, and failure handling to ensure system resilience and robust data ingestion across distributed environments.

•ChatApp Backend

2023

Scalable real-time messaging architecture with AI integration and WebSocket communications.

🐙 GitHub

- Designed scalable real-time backend architecture using WebSockets, Celery, and RabbitMQ, implementing message queue-based communication for 20,000+ daily active requests.
- Implemented JWT authentication, secure WebSocket handshakes, and database indexing strategies to optimize state-aware real-time messaging performance and security.

EDUCATION

•Bachelor of Technology in Computer Science and Engineering

2021 – 2025

Techno College of Engineering Agartala

TECHNICAL SKILLS

Core Backend: Microservices Architecture, RESTful API Design, System Design, SOLID Principles, Design Patterns
Cloud & DevOps: CI/CD, GitHub Actions, Docker, Containerization, Nginx, Reverse Proxy, Load Balancing, AWS/GCP
Security & Testing: JWT, OAuth2, RBAC, PyTest, Integration Testing, API Security Best Practices, Rate Limiting
Frameworks & Tools: FastAPI, Node.js, WebSockets, Celery, RabbitMQ, Redis, OpenAPI/Swagger, SQLAlchemy
Languages & Databases: Python, JavaScript, SQL, Solidity, PostgreSQL, MongoDB, SQLite